3d Computer Vision

§5 Registration

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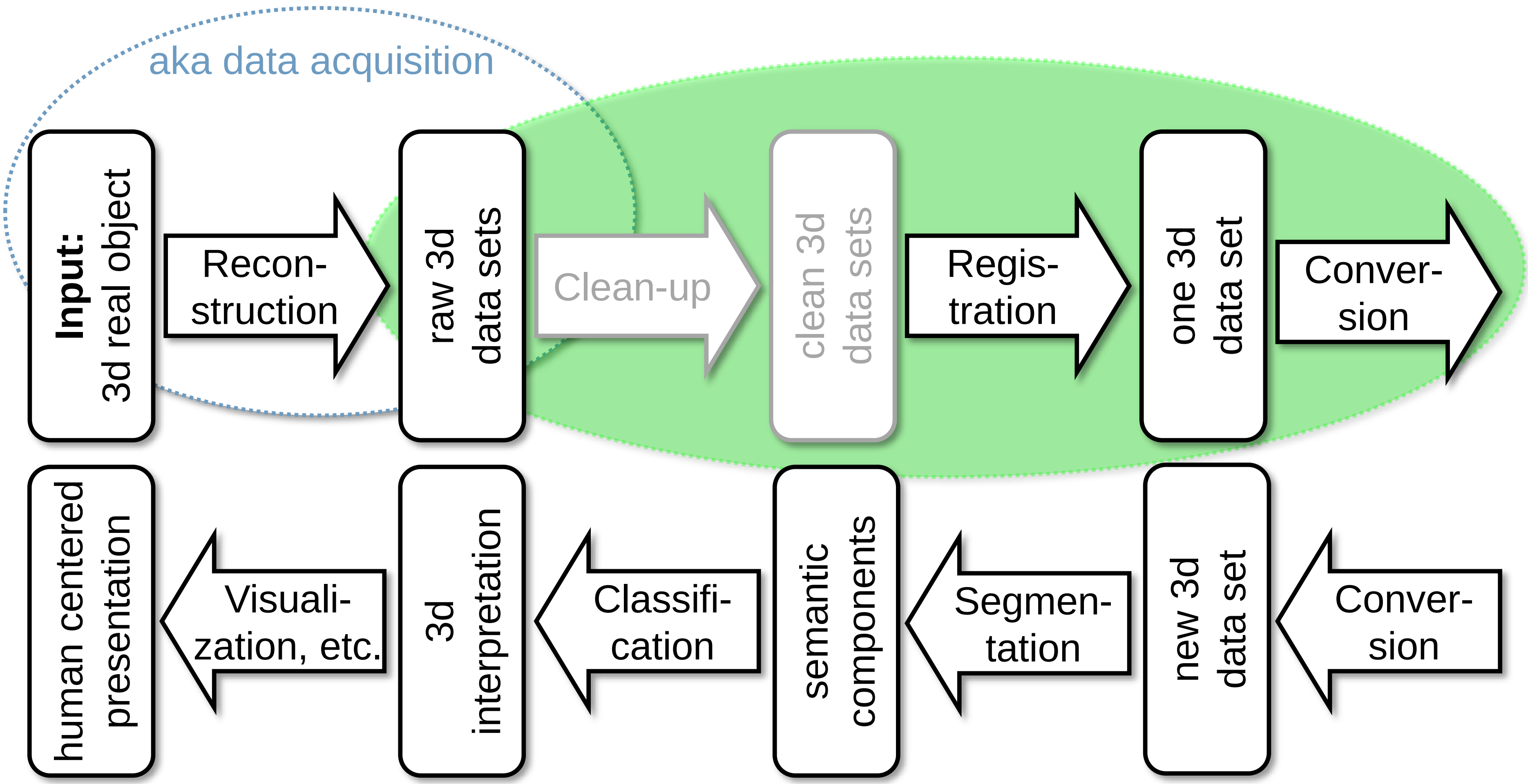
Content

1. Overview

3d Reconstruction Pipeline



1. Overview

Registration

- Application and function in the reconstruction pipeline:
  - Merge partial 3d point-clouds from different scans to yield a complete 3d point-cloud.
  - Align 3d point-clouds for comparison, aka shape matching.
- Usually a two-stage process
  1. **Coarse registration** (aka pre-registration) to match poses of point clouds roughly, i.e., minimize a global error.
  2. **(Fine) registration** to match poses of point-clouds as exact as possible, i.e., minimize a local error.

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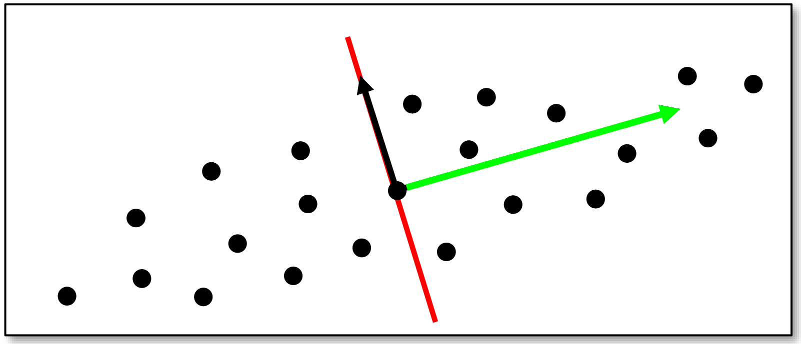
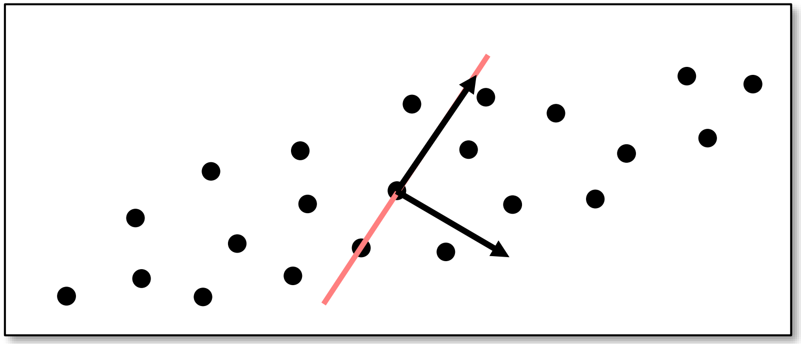
1. Overview
2. Coarse registration, aka pre-alignment
   a. Principal Component Analysis, aka PCA
   b. RANdom SAmple Consensus, aka RANSAC
   c. 4-Points Congruent Sets, aka 4PCS
3. Fine registration, aka alignment
   a. Iterative Closest Points, aka ICP

Principal Component Analysis (PCA)

- Given a point-cloud $\{p_1, \ldots, p_m\} \subset \mathbb{R}^3$.
- How to choose the hyper-planes for the partitions in a BSP-tree?
- What is the direction with maximal (minimal) diameter of the point-cloud?
- What is the best fitting plane to the point-cloud?
- The PCA yields an orthogonal coordinate system e_1, e_2, e_3, which is aligned with the point-cloud.



or



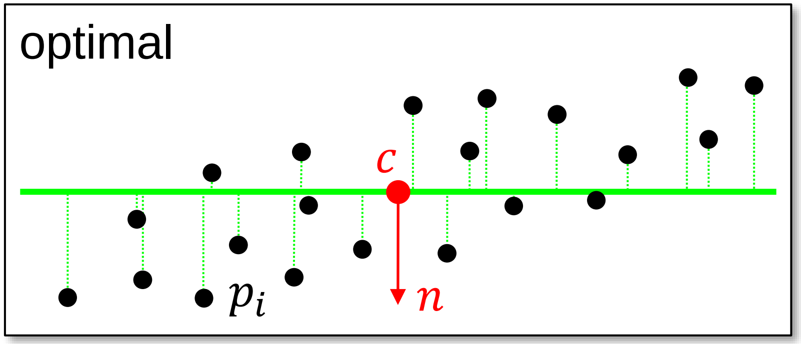
2a. Principal component analysis (PCA)

- A plane P is defined by a (normalized) normal n and a point c in the plane via

$$P = \{p \in \mathbb{R}^3 \mid 0 = n^t(p - c) =: d_P(p)\}.$$

- Determine n and c to minimize the mean squared point-plane-distances

$$D_P = \frac{1}{m}\sum d_P^2(p_i).$$

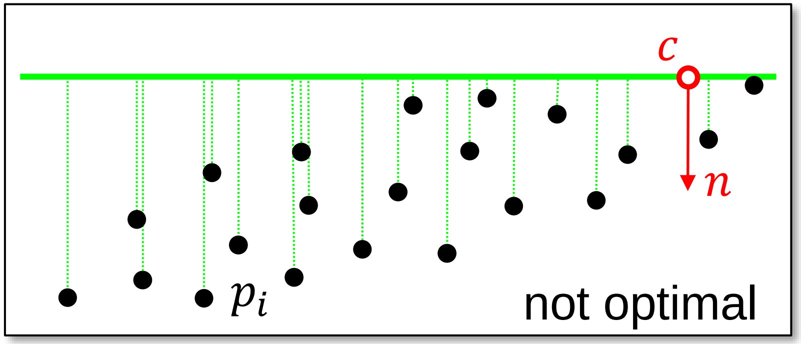


(1) How to choose c?

- For arbitrary normal n, the plane located at the barycenter of the points

$$c = \frac{1}{m}\sum_{i=1}^{m} p_i$$

minimizes D_P.



2a. Principal component analysis (PCA)

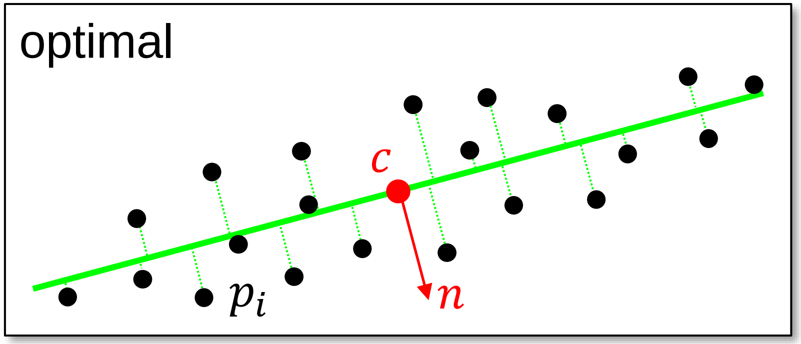
(2) How to choose n?

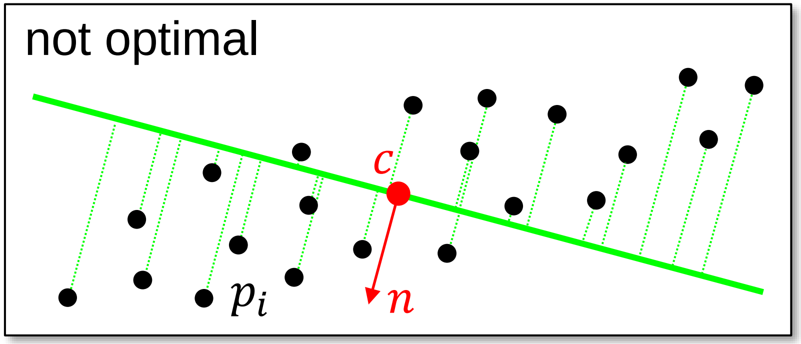
- Taking for the normal n the eigenvector e_3 corresponding to the smallest eigenvalue λ_3 of

$$M = A^t A \in \mathbb{R}^{3\times 3} \quad \text{with}$$

$$A = \begin{pmatrix} (p_1 - c)^t \\ (p_2 - c)^t \\ \vdots \\ (p_m - c)^t \end{pmatrix}$$

minimizes D_P.

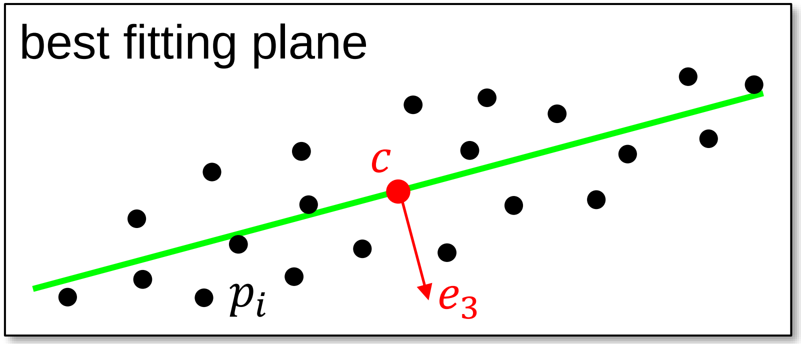


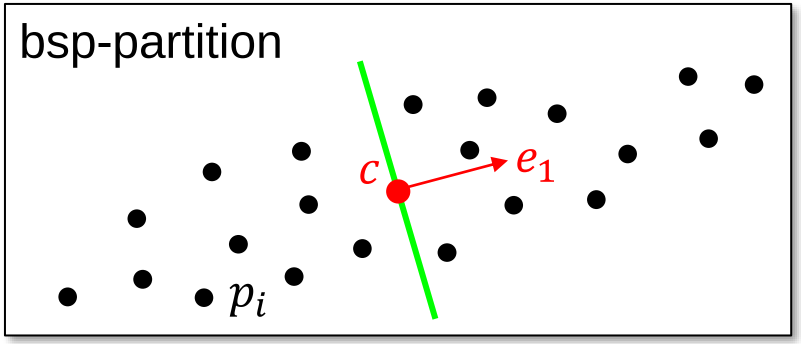


- The matrix M is the so-called covariance matrix.

2a. Principal component analysis (PCA)

- The matrix M is symmetric, positive semi-definite.
  - It has real eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ with $\lambda_k e_k = M e_k$ and
  - pairwise orthogonal eigenvectors e_1, e_2, e_3, i.e., $e_k \perp e_\ell, k \neq \ell$,
  - for $\|e_1\| = \|e_2\| = \|e_3\| = 1$.
- Stretch of the point-cloud in direction e_k is proportional to $\sqrt{\lambda_k}$, $k = 1,2,3$.
- The **best fitting plane** is the plane defined by the barycenter c and normal e_3.
- The **best plane for the bsp-tree partition** is defined by the bary-center c and normal e_1,
  - aka best splitting plane.

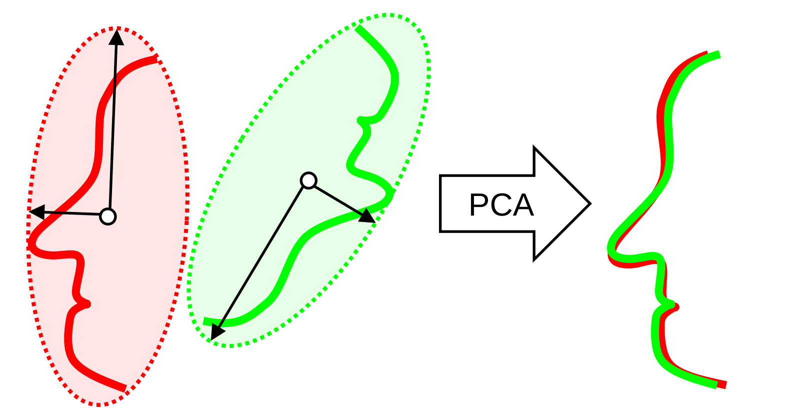




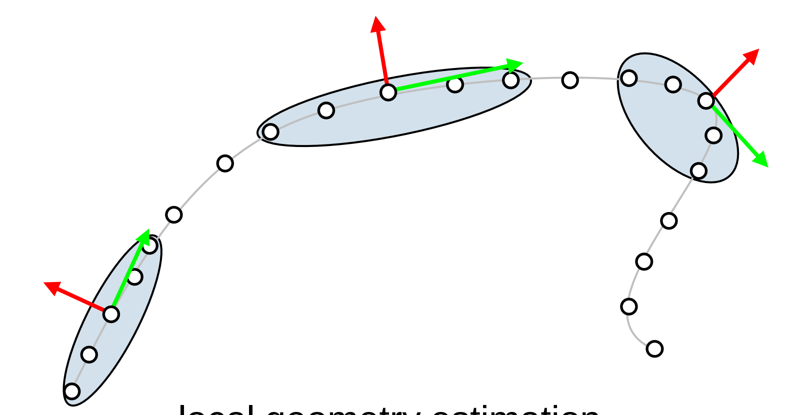
2a. Principal component analysis (PCA)

Remarks

- Analog in arbitrary dimensions.
- Further applications:
  - pre-alignment,
  - local geometry estimation, e.g., normals or tangent planes,
  - dimension reduction,
  - de-noising,
  - data compression,
  - model completion,
  - etc.



pre-alignment



local geometry estimation

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   a. Iterative closest points, aka ICP

2c. Random sample consensus (RANSAC)

Input: Two 3d meshes or point-clouds P and Q.

Output: Rigid body motion (rotation R, translation T) mapping P onto Q.

- PCA yields approximate transformations only due to noise.
- PCA works robustly only, if both point clouds are "similar in shape".

Recap: Three points on P suffice to define a local Cartesian system S_1.

2c. Random sample consensus (RANSAC)

Input: Two 3d meshes or point-clouds P and Q.

Output: Rigid body motion (rotation R, translation T) mapping P onto Q.

- PCA yields approximate transformations only due to noise.
- PCA works robustly only, if both point clouds are "similar in shape".

Idea: 1. Three points on P suffice to define a local Cartesian system S_1.

2. The corresponding point-triple on Q yields a corresponding local Cartesian system S_2.
3. Compute rigid body motion R, T mapping S_1 to S_2.

Question: How to choose corresponding point-triples?

- Exhaustive search for point correspondences?
- ► Take a randomized approach…

Answer: **RANdom**ly **SAmple** and decide for the rigid body motion with the largest **Consensus** and iterate, aka RANSAC.

2c. Random sample consensus (RANSAC)

Random sample consensus (RANSAC)
Input: Two point-clouds $P \subseteq \mathbb{R}^3$ and $Q \subseteq \mathbb{R}^3$. **Output:** Relative rotation R and translation T.
(1) **Select** multiple, random, non-collinear point-triples $(p_1, p_2, p_3)_{i=1,\dots,m,} \in P^3$ and $(q_1, q_2, q_3)_{j=1,\dots,n} \in Q^3$.
(2) **Compute** the rigid body motion T_{ij} between each point-triple pair for $i = 1, \dots, m$ and $j = 1, \dots, n$.
(3) **Consensus:** Decide for the rigid body motion T_{ij}, that yields the maximum number of correspondences or the minimal distance between P and Q after alignment with T_{ij}.
(4) **Align** the point sets using T_{ij}.
(5) **Iterate** steps (1)-(4) until satisfied…

2c. Random sample consensus (RANSAC)

More efficient variant DARCES (data-aligned rigidity-constrained exhaustive search)

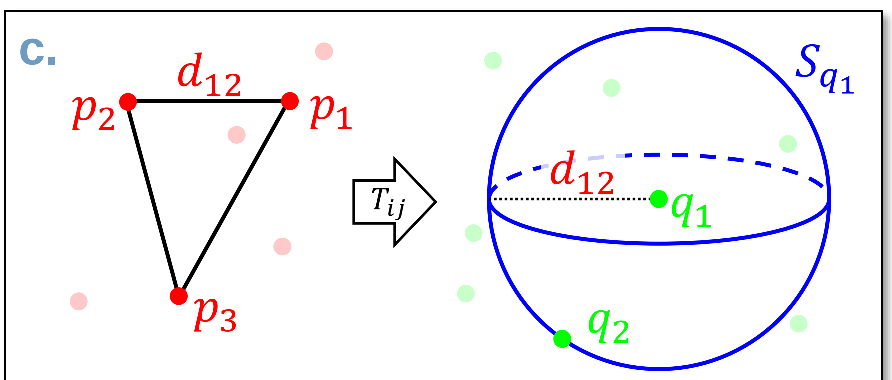
- Improve the “random” selection to speed up correspondence-search…

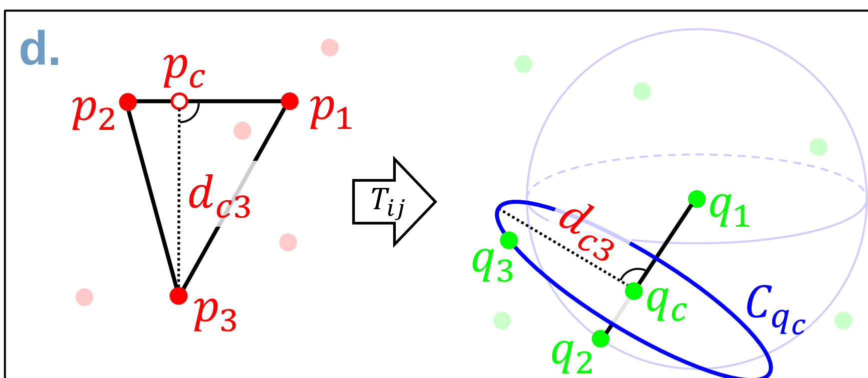
(1) a. Choose random point-triple $(p_1, p_2, p_3) \in P^3$ first.

b. Choose random point $q_1 \in Q$.

c. Choose point $q_2 \in Q$ on sphere S_{q_1} at q_1 with radius $\|p_1 - p_2\| = d_{12}$.

d. Choose point $q_3 \in Q$ on circle C_{q_c} at q_c with radius $\|p_c - p_3\| = d_{c3}$ perpendicular to $\overline{q_1 q_2}$, where p_c is the orthogonal projection of p_3 onto $\overline{p_1 p_2}$ and q_c is the corresponding (i.e., same ratio) point on $\overline{q_1 q_2}$.





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2b. 4-Points congruent sets (4PCS)

Input: Two 3d point-clouds P and Q.

Output: Rotation R and translation T mapping P onto Q.

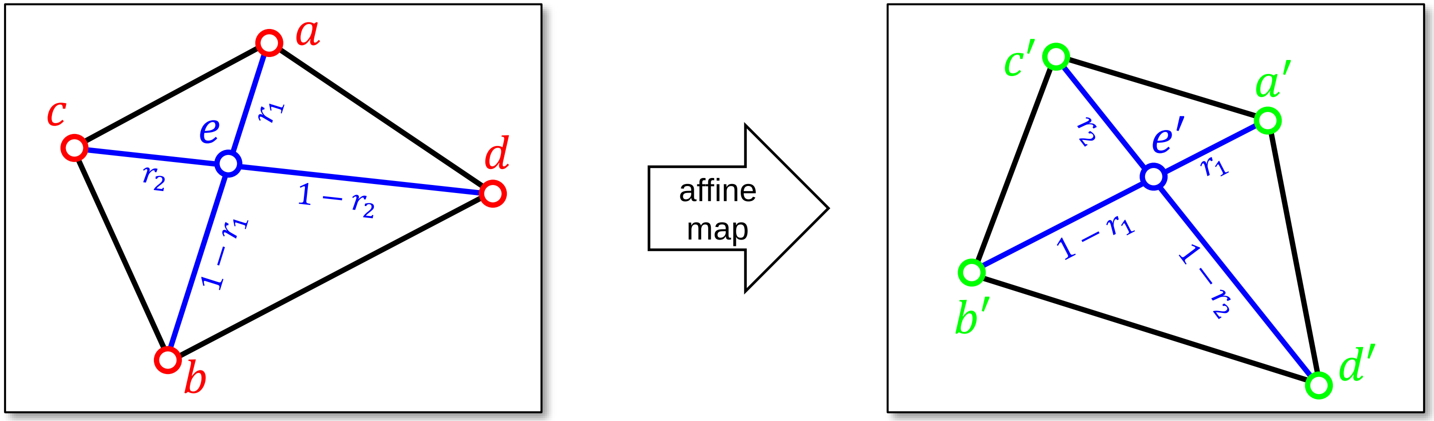
The same requirements as before.

Recap: Ratios are invariant with respect to affine maps.

Idea: Choosing four coplanar, non-collinear points $a, b, c, d \in P$, the diagonals of the quadrangle intersect in a point e with ratio

$$r_1 = \frac{\|a-e\|}{\|a-b\|} \text{ and } r_2 = \frac{\|c-e\|}{\|c-d\|}.$$

- The corresponding points $a', b', c', d' \in Q$ will yield the same ratios.



2b. 4-Points congruent sets (4PCS)

- If four coplanar (non-collinear) points $p_1, \dots, p_4 \in P$, with ratios r_1, r_2 are picked, determine the corresponding points in Q as follows:

picking congruent points

- For a first pair of points $q_1, q_2 \in Q$ the corresponding point e_1' is
$$e_1' = q_1 + r_1(q_2 - q_1) \text{ or } e_1' = q_2 + r_1(q_1 - q_2).$$
- For a second pair of points $q_3, q_4 \in Q$ the corresponding point e_2' is
$$e_2' = q_3 + r_2(q_4 - q_3) \text{ or } e_2' = q_4 + r_2(q_3 - q_4).$$
- Only if the intermediate points e_1', e_2' coincide, the points q_i are (probably) a rigid transformation of the p_i,
  - aka the quadruples are congruent.

- Use these congruent quadruples of points to estimate the rotation R and translation T.

2b. 4-Points congruent sets (4PCS)

4-points congruent sets (4PCS)
Input: Two point-clouds $P \subseteq \mathbb{R}^3$ and $Q \subseteq \mathbb{R}^3$. **Output:** Relative rotation R and translation T.
(1) **Select** three random points $p_1, p_2, p_3 \in P$ and pick a fourth point $p_4 \in P$, such that the quadruple is "wide" and approximately coplanar. (2) **Compute** the ratios r_1, r_2. (3) **Picking congruent points:** Extract set of congruent quadruples of points on Q, as on the previous page. (4) Compute a rigid body motion for each quadruple on Q and choose the one that yields best correspondences between P and Q. (5) **Iterate** steps (1)-(4) until satisfied…

2b. 4-Points congruent sets (4PCS)

- For a "wide" quadruple the points are
  - *sufficiently far away* to increase stability,
  - but *not too far away* to balance against sensitivity to non-overlap.

- The bottle-neck in this approach is step **(3)**.
  - In principle, it has runtime $O(m)$, if Q has m points.
  - Speed this up, using the known distances $d_1 = ||p_1 - p_2||$ and $d_2 = ||p_3 - p_4||$, i.e.
    - pick only point pairs on Q within these distances (analog to DARCES).
    - This requires a suitable data-structure, e.g., *k*d- or BSP-trees.

Evaluation

AIT/IMS

All others

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3. Fine registration, aka alignment

- Align two (partially-overlapping) 3d point clouds or meshes P and Q.
- **Application:** merging, matching, or comparing point sets.
- **Rule-of-thumb:** 60% overlap usually suffices…

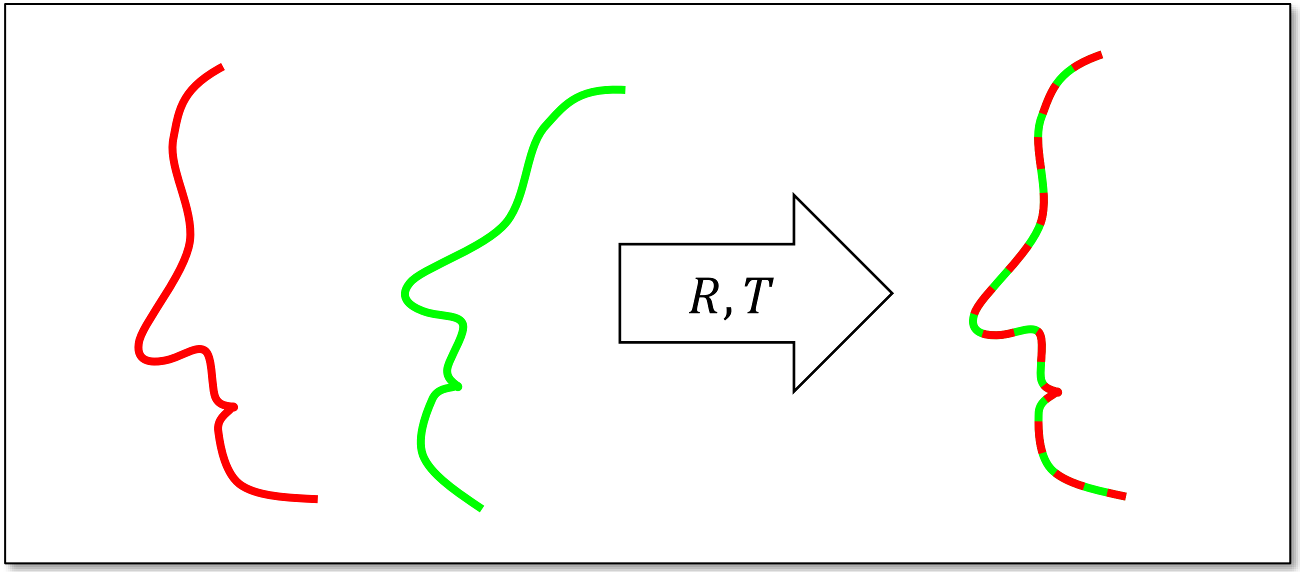
Central Question

What is the **best** transformation mapping P onto Q?

- What kind of transformation do we consider?
  - Affine? Rigid?
- How do we measure “best”?

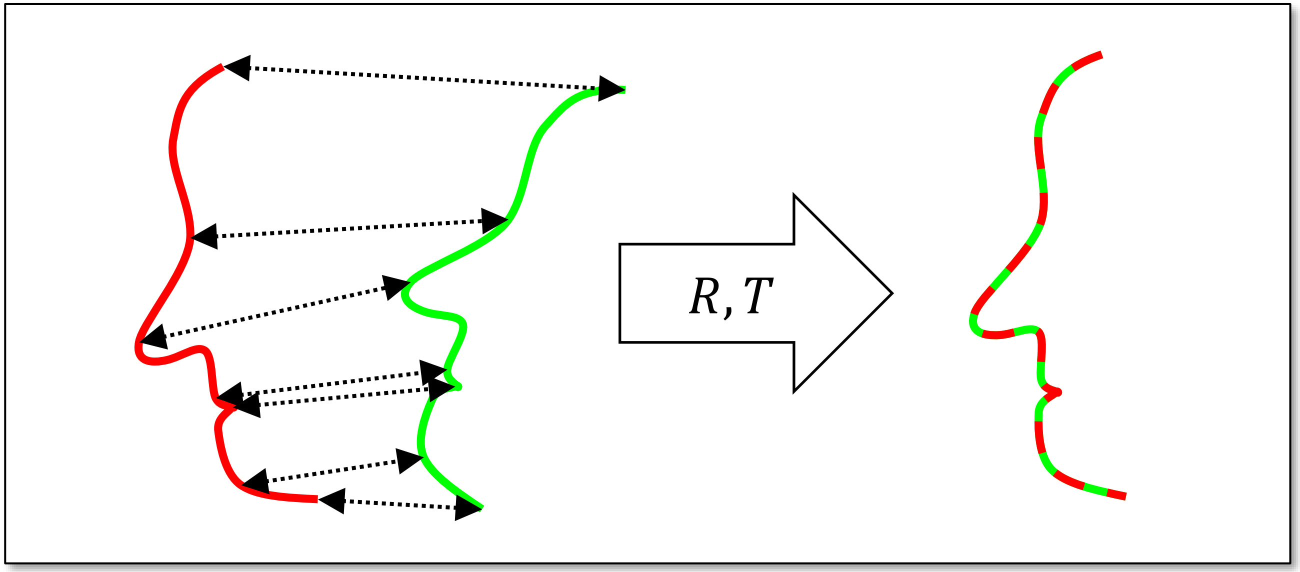
3a. Iterative closest points (ICP)

- Align two (partially-overlapping) 3d point clouds or meshes…
  - given an initial guess for the relative affine transformation.
- **Input:** Two coarsely aligned 3d point clouds P and Q.
- **Output:** Rotation R and translation T for “perfect” alignment.



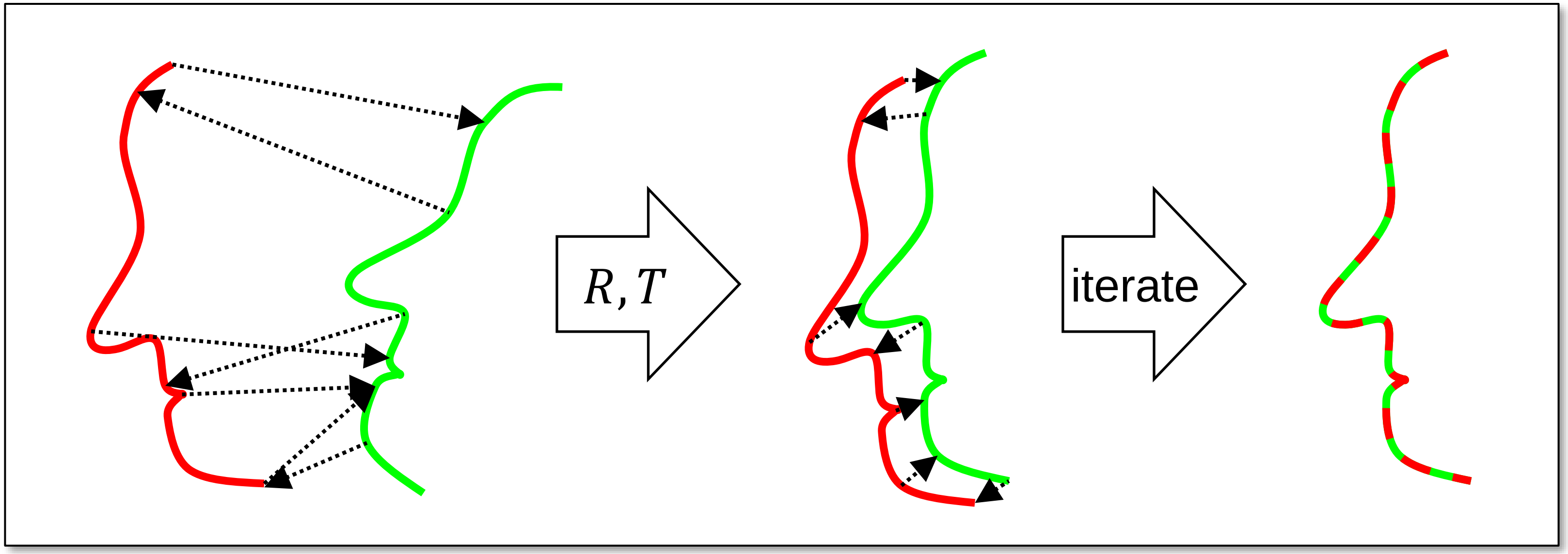
3a. Iterative closest points (ICP)

- If correct correspondences are known, problem reduces to compute R and T.
- But how to find correct correspondences?
  - User input
  - Feature detection



3a. Iterative closest points (ICP)

- **Assumption:** If the point clouds a coarsely aligned, closest points correspond to each other.
  - Choose random samples in both point clouds and compute R and T.
- Iterate to improve the point-wise error.
- This approach converges, if the initial poses are “close enough”.



3a. Iterative closest points (ICP)

Input: Two roughly aligned point clouds

$$P = \{p_i\}_{i=1}^{M} \subseteq \mathbb{R}^3 \text{ and } Q = \{q_j\}_{j=1}^{N} \subseteq \mathbb{R}^3.$$

(1) **Select** random points $\{p_{i_k}\}_{k=1}^{m} =: P_m \subset P$ and $\{q_{j_l}\}_{l=1}^{n} =: Q_n \subset Q$ with $m \ll M$ and $n \ll N$.

(2) **Match** each point from $P_m \cap Q_n$ to closest point in the other point cloud.

- This yields point-pairs

  $C = \{(p, q) \mid P_m \ni p_{i_k} = p \to q \in Q \text{ or } Q_n \ni q_{j_l} = q \to p \in P\}.$

(3) **Reject** point pairs with distance much larger than median distance.

(4) **Compute** rotation R and translation T **minimizing** the error function

$$E(R, T) = \sum_{(p,q)\in C} \|R \cdot p + T - q\|^2.$$

Output: Relative rotation R and translation T.

"$\to$" = match-to operator

3a. Iterative closest points (ICP)

Iterative closest points (ICP)

Input: Two coarse aligned point-clouds $P = \{p_i\}_{i=1}^{M}, Q = \{q_j\}_{j=1}^{N} \subseteq \mathbb{R}^3$.

Output: Relative rotation R and translation T.

(1) **Select** random points $\{p_{i_k}\}_{k=1}^{m}$ and $\{q_{j_l}\}_{l=1}^{n}$ for $m \ll M$ and $n \ll N$.

(2) **Match** every selected point to the closest point in the other point cloud, yielding a set of point-pairs

$$C = \{(p, q) |\ p = p_{i_k} \to q \in Q \vee q = q_{j_l} \to p \in P\} \subset P \times Q \,.$$

(3) **Reject** point pairs with distance much larger than median distance and remove them from C, aka clean-up.

(4) **Compute** R and T **minimizing** the error function on C

$$E = \sum_{(p,q)\in C} \|R \cdot p + T - q\|^2 \,.$$

Translation

- Compute barycenters

$$\bar{p} = \frac{1}{|C|} \sum_{(p,q)\in C} p\,, \qquad \bar{q} = \frac{1}{|C|} \sum_{(p,q)\in C} q \quad .$$

- Optimal translation is given by $T = \bar{q} - R\bar{p}$.

▶ Re-center points, i.e., $\hat{p}_i = p_i - \bar{p}$ and $\hat{q}_j = q_j - \bar{q}$.

▶ Minimize

$$\hat{E}(R) = \sum_{(p,q)\in C} \|R \cdot \hat{p} - \hat{q}\|^2 \,.$$

3a. Iterative closest points (ICP)

Rotation

- Minimize $\hat{E}(R) = \sum_{(p,q)\in C} ||R \cdot \hat{p} - \hat{q}||^2$ in terms of $R \in SO(3)$.
- Approximate rotation $R \in SO(3)$ by general linear map $A \in \mathbb{R}^{3\times 3}$

$$\min_{R\in SO(3)} \sum_{(p,q)\in C} ||R \cdot \hat{p} - \hat{q}||^2 \xrightarrow{\text{solve instead}} \min_{A\in\mathbb{R}^{3\times 3}} \sum_{(p,q)\in C} ||A \cdot \hat{p} - \hat{q}||^2$$

- The least squares solution linear map in $\mathbb{R}^{3\times 3}$ is given by

$$A = \left(\sum_{(p,q)\in C} \hat{q} \cdot \hat{p}^t\right)\left(\sum_{(p,q)\in C} \hat{p} \cdot \hat{p}^t\right)^{-1}.$$

- Compute the singular value decomposition (SVD) of A and use it to compute R via the polar decomposition, i.e.

3a. Iterative closest points (ICP)

Rotation

- Minimize $\hat{E}(R) = \sum_{(p,q)\in C} \|R \cdot \hat{p} - \hat{q}\|^2$ in terms of $R \in SO(3)$.
- Approximate rotation $R \in SO(3)$ by general linear map $A \in \mathbb{R}^{3\times 3}$

$$\min_{R\in SO(3)} \sum_{(p,q)\in C} \|R \cdot \hat{p} - \hat{q}\|^2 \xrightarrow{\text{solve instead}} \min_{A\in \mathbb{R}^{3\times 3}} \sum_{(p,q)\in C} \|A \cdot \hat{p} - \hat{q}\|^2$$

- …
- Compute the singular value decomposition (SVD) of A and use it to compute $\boldsymbol{R = U \cdot V^t \in SO(3)}$ via the polar decomposition, i.e.

$$A = \underbrace{U}_{\in SO(3)} \cdot \underbrace{\Sigma}_{\in \mathbb{R}^{3\times 3}} \cdot \underbrace{V^t}_{\in SO(3)},$$

$$\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \sigma_3), \qquad \sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0.$$

3a. Iterative closest points (ICP)

Variants of the ICP-algorithm

Ad (1) Select: **a.** All points vs. uniform subsamples vs. random samples…

b. Select points with *uniform distribution of normals*[1].

Ad (2) Match: **a.** Use a *BSP-tree* or *kd-tree* with median-partitioning to speed up the closest-point search.

b. Use *closest compatible points*, where compatible means points with the same *color, normal*[1], or other *attributes*.

c. Use *projection-based matching*, i.e., project point onto the other point cloud along prescribed direction.

d. Use *weighted correspondences*.

Ad (4) Compute: **a.** Error metric: *point-to-plane-distance*[1] instead of point-to-point-distance improves stability.

[1] Estimate normals/tangent planes with local PCA.

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   b. There is no 3b in this chapter…;-)

Questions

- What is the difference between coarse and fine registration?
- What is the PCA, how do you compute a PCA, and where did we use a PCA in the lecture?
- How do you estimate point-normals?
- What is the idea of RANSAC and how did we use it for registration?
- What is DARCES?
- What is the principle of 4PCS?
- What is the principle of ICP?